

# Jeff Brozena

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## Education

- 2021– **Ph.D., Informatics**  
The Pennsylvania State University • Advised by Dr. Saeed Abdullah  
NSF Graduate Research Fellow  
Research Interests: *Human-Computer Interaction, Mental Health Informatics, Digital Health, and Financial Technologies*
- 2019 **B.Sc., Information Sciences and Technology**  
The Pennsylvania State University  
Minor: Science, Technology, and Society Studies (STS)

## Research Experience

- 2021– **Supportive Financial Technologies for Bipolar Disorder**  
*The Wellbeing and Health Innovation Lab, The Pennsylvania State University*
- Survey Experiments**
- Factorial vignette studies assessing level of comfort with third-party financial data sharing and digital health interventions (adults with bipolar disorder, N=980)
  - MaxDiff scaling of financial intervention feature preferences (adults with bipolar disorder, N=150)
- Clinical Population Recruitment and Data Collection**
- Two continuous years of retrospective financial transaction histories per participant (adults with bipolar disorder, N=50)
  - Monthly anchored retrospective mood logs (adults with bipolar disorder, N=50)
  - Semi-structured qualitative interviews (adults with bipolar disorder, N=50)
- Time Series Modeling and Prediction**
- Application of *hmmTMB* to infer latent illness states from longitudinal financial data
  - Application of transformer-based *MedTsLLM* to detect anomalies and segment temporal regimes in longitudinal financial and mood data
  - Vector autoregression of eight years of uninterrupted daily wearable-derived HRV anomalies and self-reported mood logs, no missing data (N-of-1; adult with bipolar disorder)
- Qualitative Analysis**
- Grounded theory analysis of semi-structured interviews

## Professional Experience

- 2011–2017 **Director of Operations**  
*Smiles 4 Keeps Pediatric Dentistry*
- Led an end-to-end digital transformation at one of the country's largest privately-held multi-site Medicaid pediatric dental groups **while it continued to serve over 40,000 children annually**. Digitized all clinical and financial systems, trained 100+ staff on the redesigned environment, and rendered the business legible to third parties during due diligence phases. My work afforded Smiles 4 Keeps an accurate third-party valuation and **resulted in the sale of the business**. Invited to present to 5,000 pediatric dentists at the American Academy of Pediatric Dentistry conference on the subjects of digitization and compliant record-keeping.

## Skills

**Research Methods:** Experimental Research, Survey Experiments (Factorial Vignettes, MaxDiff), Mixed-Methods Research, Human Subjects Research, Clinical Population Recruitment, Longitudinal Studies, Anchored Retrospective Recall, Wearable Sensing, Interviews

**Data Analyses Techniques:** Advanced Applied Statistics, Multilevel Modeling (*lme4*), Applied Time-Series Modeling, Multilevel Bayesian Hidden Markov Modeling (*hmmTMB*), Marginal Effects Estimation (*marginaleffects*), Grounded Theory for Qualitative Data

**Programming Languages, Tools, and Frameworks:** R, Python, Latex, Django, PostgreSQL, RESTful APIs, Figma

## Publications

### Preprints

- 2025 **Jeff Brozena**, Johnna Blair, Dahlia Mukherjee, Erika F. H. Saunders, Thomas Richardson, Mark Matthews, Saeed Abdullah. The Currency of Mood: Assessing Acceptance and Privacy Preferences of Third-party Financial Data Sharing in Bipolar Disorder. In submission. <https://doi.org/10.31234/osf.io/syrwu.v1>.
- 2023 **Jeff Brozena**. Vector Autoregression (VAR) of Longitudinal Sleep and Self-report Mood Data. <https://doi.org/10.48550/arXiv.2510.02511>.

### Conference Papers

- 2024 **Jeff Brozena**, Johnna Blair, Thomas Richardson, Mark Matthews, Dahlia Mukherjee, Erika F. H. Saunders, Saeed Abdullah. 2024. Supportive Fintech for Individuals with Bipolar Disorder: Financial Data Sharing Preferences to Support Longitudinal Care Management. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA. <https://doi.org/10.1145/3613904.3642645>.

### Journal Articles

- 2025 Emma Morton, Elsy Willis, **Jeff Brozena**, Andrew Kcomt, Erin E. Michalak. 2025. The Type, Impacts, and Experiences of Peer Support for People Living With Bipolar Disorder: A Scoping Review. *Bipolar Disorders*. <https://doi.org/10.1111/bdi.70006>.
- 2022 Johnna Blair, **Jeff Brozena**, Mark Matthews, Thomas Richardson, Saeed Abdullah. Financial technologies (FinTech) for mental health: The potential of objective financial data to better understand the relationships between financial behavior and mental health. *Front. Psychiatry* 13:810057. <https://doi.org/10.3389/fpsyt.2022.810057>.

### Presentations

- 2026 **Jeff Brozena**, Johnna Blair, Dahlia Mukherjee, Erika F.H. Saunders, M.D., Saeed Abdullah. 2026. Mapping Financial Behavior in Bipolar Disorder: Development of a Dataset Linking Spending Habits to Mood Fluctuations. To be presented at the 28th Annual Conference of the International Society for Bipolar Disorders, Vancouver, Canada, September 2026; Poster presented at Penn State College of Medicine Psychiatry and Behavioral Health Research Conference, Hershey, PA, April 2026. **Awarded Exemplary Graduate Student Research Poster.**
- 2026 **Jeff Brozena**, Saeed Abdullah. Evidence-based digital design: Utilizing MaxDiff findings to guide the development of financial interventions in bipolar disorder. To be presented at the 28th Annual Conference of the International Society for Bipolar Disorders, Vancouver, Canada, September 2026.
- 2026 **Jeff Brozena**, Saeed Abdullah, Lan Kong, Erika F.H. Saunders, Guodong Liu. Eight Years of Autonomic Monitoring: An N-of-1 Longitudinal Study of Wearable-derived HRV Anomalies and Self-reported Mood Logs. To be presented at the 28th Annual Conference of the International Society for Bipolar Disorders, Vancouver, Canada, September 2026.

### Workshops

- 2026 Chris Elsdén, Belen Barros Pena, Helena Lyhme, **Jeff Brozena**, Daniel Mwesigwa, Chris Speed, Jofish Kaye, John Vines. Configuring Money as an Interface. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI EA '26)*. Association for Computing Machinery, New York, NY, USA. <https://moneyasinterface.designinformatics.org/workshop.paper.pdf>.
- 2025 Johnna Blair, **Jeff Brozena**, John Vines, Jofish Kaye, Mark Matthews, Saeed Abdullah. 2025. Future of Money and HCI. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*. Association for Computing Machinery, New York, NY, USA, Article 780, 1–6. <https://doi.org/10.1145/3706599.3706711>.

### Awards and Scholarships

- 2025 Sawtooth Software Academic Grant
- 2023 NSF Graduate Research Fellowship
- 2011 Rock Ethics Award, Pennsylvania State University

## Professional Affiliations

- 2023 [Association for Computing Machinery](#)
- 2023 [CrestBD Network Member](#)
- 2022 [Society for Digital Mental Health](#)

## Teaching

- 2022 Teaching assistant, Methods for Studying Users (HCDD 364w)
- 2022 Teaching assistant, Mobile Application Design and Development (HCDD 340)
- 2021 Teaching assistant, Application Development Design Studio (IST 261)